

## **PROJECT 2**

### **CHINOOK DATABASE SQL ANALYSIS REPORT**

#### **Project Overview**

This project was completed using Microsoft SQL Server and the Chinook Database to demonstrate practical SQL skills in data extraction, transformation, querying, and business analysis. The Chinook database represents a digital music store containing customer purchases, invoices, artists, albums, tracks, playlists, and media types.

The project focused on applying SQL concepts such as filtering, aggregation, joins, and window functions to solve real-world business problems and generate actionable insights.

#### **Database Setup**

The database consisted of multiple related tables including Customer, Invoice, InvoiceLine, Track, Album, Artist, Genre, Playlist, and MediaType. Relationships between tables were already established using primary and foreign keys, which enabled efficient relational analysis and data retrieval.

An Entity Relationship Diagram (ERD) was also used to visualize table relationships and understand how customer purchases connect to tracks, albums, and artists.

#### **SQL Analysis & Insights**

##### **A. Customer Analysis**

A total of **59 customers** across **24 countries** were analyzed. The **USA** recorded the highest number of customers, indicating it as the business's largest customer market.

## **Top Customers by Revenue**

The highest-spending customer was **Helena Holý** from the Czech Republic with a total spending of **\$49.62**, followed closely by customers from the USA, Chile, Ireland, and Hungary. This analysis helped identify high-value customers important for customer retention strategies.

### **B. Revenue Analysis**

The total business revenue generated was **\$2,328**, while the average invoice value was **\$5.65**, indicating moderate customer spending per transaction.

### **Revenue by Country**

The **USA** generated the highest revenue at **\$523.06**, followed by:

- Canada — **\$303.96**
- France — **\$195.10**
- Brazil — **\$190.10**
- Germany — **\$156.48**

These insights revealed the company's strongest performing markets and regions contributing most to overall sales.

### **C. Artist & Album Analysis**

The database contained **347 albums**.

### **Artists with Multiple Albums**

The most productive artist was **Iron Maiden** with **21 albums**, followed by:

- Led Zeppelin — 14 albums
- Deep Purple — 11 albums
- Metallica — 10 albums
- U2 — 10 albums

### **Album with the Highest Tracks**

The album **Greatest Hits** contained the highest number of tracks with **57 songs**, indicating extensive music collections within the catalog.

### **D. Genre & Media Analysis**

#### **Most Popular Genre**

The most purchased genre was **Rock** with **835 purchases**, significantly outperforming:

- Latin — 386 purchases
- Metal — 264 purchases
- Alternative & Punk — 244 purchases

This showed a strong customer preference for rock music.

#### **Most Used Media Type**

The most common media format was **MPEG audio file** with **3,034 tracks**, demonstrating its dominance within the music store inventory.

### **E. Playlist & Track Analysis**

#### **Playlist with the Most Tracks**

The playlist **Music** contained the highest number of tracks with **6,580 songs**, followed by **90's Music** with **1,477 tracks**.

#### **Tracks Generating the Most Revenue**

The tracks **Dazed and Confused** and **The Trooper** generated the highest revenue at **\$4.95** each, making them the top-performing songs in the database.

## **SQL Techniques Applied**

The project utilized several SQL concepts including:

- SELECT, WHERE, ORDER BY
- GROUP BY and HAVING
- Aggregate functions (SUM, AVG, COUNT)
- INNER JOIN operations
- Window functions such as RANK()

## **Conclusion**

This project demonstrated the use of SQL Server for solving business problems through structured data analysis and relational querying. The analysis provided valuable insights into customer behavior, sales performance, revenue trends, music preferences, and artist productivity.

Overall, the project strengthened practical SQL querying skills, improved understanding of relational databases, and enhanced the ability to transform raw data into meaningful business intelligence suitable for decision-making.

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